WINNER RANKINGS

1	Horseshoe False Discovery Rate Control via Frequentist Thresholding Im	65%
2	T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co-	63%
3	Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Ca	62%
4	GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyon	62%
5	**Conditional Independence Masking in JEPA Targets Suppresses Co-express ◀**	**60%**

Conditional Independence Masking in JEPA Targets Suppresses Co-expression Shortcuts in GRN Initialization

Status: Winner · **Domain:** Gene Regulatory Network Inference / Causal Representation Learning · **Generation:** 0 · **Score:** 60%

Abstract

The Stage 1 co-expression problem in two-stage GRN inference arises because JEPA encoders trained with MSE loss cannot distinguish direct regulatory interactions from indirect co-expression mediated by shared regulators or latent confounders. Literature establishes that MSE convergence across structurally distinct networks masks the identification of unique causal structures [21] , and that causal GRN methods require conditional independence constraints to break the equivalence between causal and spurious correlational structures [59, 38]. We hypothesize that augmenting the Stage 1 JEPA prediction target with a conditional independence penalty—specifically, penalizing W_init entries where the partial correlation between TF i and gene j, conditioned on all other TFs, is low—will selectively suppress co-expression-driven weights and increase the causal specificity of W_init. Unlike standard JEPA which treats all predicted weights equally, this conditional masking re-weights the MSE loss by partial correlation strength, effectively teaching the encoder to distinguish hub co-expression from direct regulation. On DREAM5 net3, where 334 TFs create a high-dimensional conditioning problem, we propose using a sparse graphical model (graphical horseshoe) to estimate partial correlations efficiently. Expected outcomes include W_init AUROC ≥ 0.62 on net3 and AUPR > 0.099 in Stage 2, with net2 AUPR maintained above 0.0164.

1. INTRODUCTION & SIGNIFICANCE

Paper [21] demonstrates that even with high-quality data, many mechanistic models of gene regulation are practically indistinguishable under MSE or likelihood objectives—the inverse problem is fundamentally underdetermined, and loss convergence does not imply causal identification. Paper [59] establishes that dropout-corrected causal GRN inference requires conditioning on other variables to separate direct from indirect regulation. The critical gap is that no existing JEPA initialization method for GRN inference uses conditional independence structure to re-weight training targets, leaving encoders free to exploit co-expression shortcuts.

2. HYPOTHESIS STATEMENT

> *Weighting the Stage 1 JEPA MSE loss by precomputed partial correlations (conditioned on all other TFs via a graphical horseshoe estimate) will suppress co-expression-driven W_init entries and increase W_init AUROC on held-out regulatory edges by at least 0.04 AUROC units on net3.*

3. BACKGROUND & LITERATURE REVIEW

- **Practical indistinguishability in a gene regulatory network inference problem, a case study** — FitzGerald et al. (2025)
- **Gene Regulatory Network Inference in the Presence of Dropouts: a Causal View** — Dai et al. (2024)
- **Gene Regulatory Network Inference in the Presence of Selection Bias and Latent Confounders** — Luo et al. (2025)
- **Censored Graphical Horseshoe: Bayesian sparse precision matrix estimation with censored and missing data** — Mai et al. (2026)
- **Concentration of a sparse Bayesian model with Horseshoe prior in estimating high-dimensional precision matrix** — Mai (2024)
- **Efficient inference of dynamic gene regulatory networks using discrete penalty** — Ravikumar et al. (2025)
- **Machine Learning Methods for Gene Regulatory Network Inference** — Hegde et al. (2025)
- **Inferring gene expression networks with hubs using a degree weighted Lasso approach** — Sulaimanov et al. (2017)
- *+ 19 additional papers — see Section 8.*

4. METHODOLOGY

Precompute a sparse partial correlation matrix for each DREAM5 network using a graphical horseshoe estimator on TF expression profiles. Use these partial correlations as sample weights in the Stage 1 JEPA MSE loss, upweighting TF-gene pairs with strong partial correlations and downweighting those whose correlation vanishes when conditioning on other TFs. Evaluate W_init AUROC and Stage 2 AUPR.

Experimental Phases

1. For each DREAM5 network, fit a graphical horseshoe model on the TF expression submatrix (334×805 for net3) to obtain a sparse partial correlation matrix Ω; this is a one-time precomputation outside the training loop.
2. Construct a weight matrix M where M[i,j] = |Ω[i,j]| + ε for TF i and gene j (using gene-TF partial correlations approximated from the TF-TF graph and TF-gene raw correlations via a shrinkage estimator).
3. Modify Stage 1 JEPA loss to: L = Σ_{i,j} M[i,j] · (W_pred[i,j] - W_target[i,j])^2, normalizing M to avoid scale artifacts.
4. Train Stage 1 for 150–300 epochs and extract W_init; compute AUROC against DREAM5 gold standard held-out edges; compare to unweighted JEPA baseline.
5. Pass W_init to Stage 2 horseshoe regression; record AUPR on all four DREAM5 networks; perform ablation using L1-regularized partial correlations (graphical lasso) instead of graphical horseshoe for M.
6. Measure Stage 1 training time to confirm it remains <3s per 10 epochs on Apple Silicon MPS, given that M is precomputed and fixed.

5. EXPECTED OUTCOMES

Predicted Impact: W_init AUROC increase of 0.04–0.08 units on net3; net3 Stage 2 AUPR increase from ~0.090 to >0.099, beating GENIE3; no increase in Stage 1 training time (M is precomputed); potential spillover improvement on net1 where indirect regulation through shared TFs is common.

Success Criteria: W_init AUROC > 0.62 on net3; Stage 2 net3 AUPR > 0.099; net2 AUPR ≥ 0.0164; graphical horseshoe M outperforms graphical lasso M by ≥0.01 AUROC on net3 W_init evaluation; Stage 1 remains <3s/10 epochs.

Null Result Interpretation: If this hypothesis is false: Rather than weighting the MSE loss by precomputed partial correlations, use the partial correlations as a soft auxiliary label in a multi-task Stage 1 objective (predict both masked expression values and pairwise partial correlation structure), separating the causal signal integration from the MSE loss weighting and avoiding instability.

6. RISK ANALYSIS & MITIGATION

Fatal Assumptions

- ☠ The graphical horseshoe partial correlation matrix accurately recovers direct TF-TF regulatory links rather than statistical artifacts in the specific sample sizes available for each

DREAM5 network — if false: weights derived from it inject systematic noise into Stage 1, actively harming W_init quality relative to uniform weighting.

- ☠ Partial correlation conditioning genuinely filters co-expression without over-suppressing true direct regulatory signal — if false: real edges mediated by shared co-activators or TF cascades are systematically downweighted, reducing recall and AUROC.

  Weakest Assumption: That precomputed partial correlations from a graphical horseshoe on TF expression profiles are reliable and informative enough in the small-sample DREAM5 regime to serve as meaningful sample weights — in high-p, low-n settings, these estimates are too noisy to confer a genuine 0.04 AUROC gain.

Key Risks

- In high-dimensional, small-sample DREAM5 settings (e.g., net1 with hundreds of TFs and limited samples), the graphical horseshoe partial correlation matrix is poorly identified; estimated partial correlations have high variance, so the derived sample weights are dominated by noise rather than signal.
- Partial correlation conditioning on all other TFs does not distinguish between indirect regulation through TF cascades and direct regulation; a TF that acts only through another TF would have its weight zeroed out, causing true indirect regulatory edges to be systematically suppressed in W_init.
- The MSE loss weighting scheme assumes that partial correlation magnitude is monotonically related to regulatory edge probability — but partial correlations can be large for structural/confounded relationships (e.g., TF auto-regulation artifacts) while being small for weak but real direct regulation.
- Circular dependency risk: Stage 2 horseshoe is initialized from W_init, which was itself biased by a Stage-2-style graphical horseshoe pre-computation; if both stages make the same systematic errors, errors compound rather than cancel.

Minimal Validation Experiment

Compute the graphical horseshoe partial correlation matrix for net3 and measure its AUROC on held-out regulatory edges directly (as a standalone predictor). Takes ~1 day. If AUROC < 0.55, the partial correlations are uninformative/noisy and weighting by them will not help; if AUROC ≥ 0.60, proceed to weighted JEPA training.

Competing Mechanisms

- ⇄ Alternative 1 (Global Shrinkage Sufficiency): The Stage 2 horseshoe prior already provides sufficient conditional independence structure through its global-local shrinkage mechanism, rendering any Stage 1 conditioning redundant — if true, adding conditional independence masking to Stage 1 would produce no statistically significant improvement in Stage 2 AUPR compared to unweighted JEPA initialization.
- ⇄ Alternative 2 (Partial Correlation Noise): In high-dimensional settings (334 TFs, 805 samples for net3), estimated partial correlations are too noisy to serve as reliable supervision signals, and the masking matrix M adds noise rather than signal — if true, the conditionally-

masked W_init would have LOWER AUROC than the unmasked baseline, showing the partial correlation estimates are misleading.

Discriminating Experiment

On net3, run Stage 1 with three conditions: (A) conditional independence masking with graphical horseshoe M, (B) unweighted JEPA, (C) masking with permuted M (same sparsity, random edge weights as negative control). Measure W_init AUROC for all three before Stage 2. If THIS hypothesis is correct: AUROC(A) > AUROC(B) > AUROC(C). If Alternative 1 is correct: Stage 2 AUPR(A) ≈ AUPR(B), even if AUROC(A) > AUROC(B). If Alternative 2 is correct: AUROC(A) < AUROC(B).

7. LIMITATIONS & ASSUMPTIONS

Key Assumptions (most → least confident)

1. Assumption [80%]: Graphical horseshoe partial correlations are estimable with acceptable noise at net3's $n=805$, $p=334$ TF dimensions — Risk: even at $n=805$, estimating 334×334 partial correlations has ~56k parameters relative to sample size; may require strong regularization.
2. Assumption [72%]: Partial correlation strength between TF i and gene j (approximated via the TF-TF graph and TF-gene correlations) is positively correlated with true regulatory edge presence — Risk: partial correlations on expression data cannot distinguish true regulation from indirect effects through unmeasured regulators.
3. Assumption [60%]: Upweighting high partial-correlation pairs during JEPA training will shift W_init toward causal edges rather than simply scaling existing co-expression weights — Risk: the encoder may learn to predict the weighted target by amplifying all high-correlation weights, not discriminating causal from non-causal ones.
4. Assumption [45%]: The improvement in W_init causal specificity from Stage 1 will survive the Stage 2 horseshoe compression without being overwritten — Risk: Stage 2 with strong global shrinkage τ may re-initialize effective weights regardless of W_init quality.

Major Assumptions — require revision if false

- ⚠ Sample sizes in DREAM5 networks are sufficient for a sparse graphical horseshoe to converge to a stable partial correlation matrix — if false: the pre-computation step requires regularization tuning per network, adding significant implementation burden.
- ⚠ The JEPA MSE loss can meaningfully incorporate continuous sample weights without destabilizing the self-supervised training dynamics — if false: the loss landscape becomes poorly conditioned and requires learning-rate / normalization adjustments.
- ⚠ Stage 1 and Stage 2 horseshoe operate on different enough representations that the circular dependency between pre-computed weights and Stage 2 initialization does not collapse both stages to the same fixed point — if false: the two-stage design collapses to a single graphical horseshoe run.

8. EVIDENCE SUMMARY

DIMENSION	SCORE	WEIGHT
EVIDENCE GROUNDING	58%	35%
Feasibility	63%	30%
Impact Potential	60%	25%
Novelty	55%	10%
Composite	**60%**	weighted

Why this won: This hypothesis directly attacks the co-expression masking problem from the causal inference angle rather than the representation learning angle, making it genuinely distinct from Hypothesis 1. The conditional independence approach is grounded in papers [21] and [59] which identify the fundamental underdetermination problem. The graphical horseshoe for computing partial correlations ties in the horseshoe literature [9, 54] in a novel way—using it as a preprocessing tool for Stage 1 rather than as Stage 2's regression prior. The weighting scheme is computationally cheap (precomputed once), feasible on MPS, and produces a measurable change in W_init AUROC.

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[3] Luo, G., Dai, H., Li, L., Gao, C., & Sun, B. (2025). *Gene Regulatory Network Inference in the Presence of Selection Bias and Latent Confounders*.

[4] Mai, T. T., & Banerjee, S. (2026). *Censored Graphical Horseshoe: Bayesian sparse precision matrix estimation with censored and missing data*.

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